The Compactor That Couldn't Learn Where to Look: Gradient Saturation, Position-Kernel Collapse, and an Unfrozen KV Compressor

Draft v0.3 — 2026-07-20 — certification re-measured at n=60 documents
Artifacts: `memsub-artifacts:/{escape, fullrecipe-telemetry, compactor-checkpoints, compa`

INVALIDATED RESULT (2026-07-21). The E1FULL certification documents overlapped training. Certification, paired-CI, task-anchored SHIP, ranking, and deployment conclusions are withdrawn pending a clean holdout retrain and rerun.

Abstract

We report a failure mode of learned KV-cache compaction in which a compactor architecture cannot learn to route: trained under its canonical configuration, the per-layer Perceiver's block-1 cross-attention kernel is a uniform positional lattice that remains **bitwise identical to its initialization** for the entire training run — at every data scale we could afford (9.5k, 19.6k, and 76k training items; effective batch 4 and true batch 32) — while the training loss falls normally. We name this *position-kernel collapse* and identify the mechanism by gradient forensics: the canonical attention logit scale ($\times d_\ell = 256$) saturates the softmax at the position-dominated identity initialization, delivering gradients to the routing projections (W_Q, W_K) **eleven to twelve orders of magnitude** smaller than to the value pathway. The compactor's apparent learning is entirely value-side: it learns what to do with positions it can never change. Controls isolate the mechanism — init-time noise on the position anchors does not escape (not a tie-breaking problem), and content-seeded initialization stays essentially frozen while paying a large loss penalty (content without gradient does not help). The cure is one constant: at $\sqrt{d_\ell}$ scale, routing gradients wake by 10^9–10^{10}, kernel entropy rises from its de-saturated starting point, all eight KV-heads despecialize from the shared lattice (cross-head Jaccard 1.000 → 0.199), the kept sets become mildly item-adaptive (1.000 → 0.821 at mid scale), and the unfrozen compactor overtakes the frozen one on the very objective that froze it (final mid-scale loss 0.219 vs 0.365; three seeds) while losing 35% fewer facts on a fixed counterfactual subsample (0.239 vs 0.370). At full recipe (3,000 steps; a pre-registered extension rule fired at 1,500) the unfrozen compactor's kernel becomes item-adaptive (cross-item Jaccard 0.550). Its historical certification cannot support a held-out comparison because the evaluation documents overlap training; the predicate, ranking, confidence-interval, and deployment interpretations are withdrawn pending the clean rerun. Downstream, the reorganization is not what a routing-becomes-legible story predicts: write-time geometry stays loss-blind even under content routing ($d_{\min}$ AUC 0.500, all twelve checkpoints and the full protocol); the question-blind expansion gate's marginal value completes its monotone decline to **zero** (72% → 22.5% → 9.0% → 0%) while the computation-reading probe ladder fully recovers (stack 0.857) and still routes below never-expand; and fact fragility proves largely content-intrinsic — 78–91% of the unfrozen instance's failure set nests inside the frozen instances' — with a minority regime-relative component. Diagnostics — a one-pass gradient-norm check and cross-attention entropy telemetry — cost minutes and would have caught the collapse at step zero.

1 Introduction

Learned compaction promises what heuristic cache eviction cannot: a compressor optimized end-to-end to preserve a frozen model's behavior [1, 3]. The recipe we study — a per-layer Perceiver trained once against the frozen base with a behavioral KL objective — reports strong results at scale. We began by reproducing it at reduced scale for a certification study [2] and found something stranger than underperformance: the compactor was not compressing at all in the intended sense. It was keeping every third token.

This paper is the anatomy of that observation. Our contributions: **(i)** *position-kernel collapse*, a failure mode in which a compactor's routing freezes at initialization while its loss improves — undetectable from the loss curve and, we show, not a scale artifact; **(ii)** the mechanism, established by direct gradient measurement: logit-scale-induced softmax saturation at identity initialization kills the routing projections' gradient specifically, with init-noise and content-seeded-init controls ruling out the alternative explanations; **(iii)** the cure — reduce the logit scale — validated from gradient wake-up through kernel mobility, head despecialization, item-adaptivity, and training-loss crossover, at three seeds; **(iv)** the consequence: a full-recipe unfrozen compactor whose two pre-registered certification predicates dissociate *and rank compressors differently*, making the choice of certification anchor — not the compressor — the decisive design question; **(v)** the downstream reorganization of routing signals under a mobile kernel, including the correction this work forced onto our companion paper's fragility claim (§8); **(vi)** two mandatory-cheap diagnostics for anyone training compactors (§9).

2 Background and setup

Compactor recipe. One Perceiver-style compactor per transformer layer (36 layers, Qwen3-4B base, frozen throughout): per KV-head banks of t=1024 learned latent queries, two pre-norm blocks with cross-attention in each, QK-normalized attention with logit scale $d_\ell = 2d = 256$, identity-style initialization (block-diagonal identity on the value/output path, zeroed content projections, position-dominated Q/K anchors), trained with a forward-KL behavioral objective on answer-side tokens against the full-context teacher; β=0, no bias channel [3].

Certification protocol. From the companion paper [2]: counterfactually filtered MCQ probes over a four-domain ∼1B-token corpus (98,355 items); two axes never blended (aggregation: teacher KL and accuracy delta over surviving content; pointwise: silent lost-fact incidents over compacted-away spans); matched retained budget vs an eviction baseline; a noise-floor row; recurrence-depth certification. Training mixture for the full recipe: 75,998 rationale-traced items after domain balancing (nominal cap 23,563/domain, short-context drops applied).

Harness provenance. The earliest multi-GPU runs did not synchronize gradients across workers (a DDP wrapping omission found by this project and documented in Appendix G): the shipped 1/40-scale instances are single-replica, effective-batch-4 artifacts. All behavioral measurements in the companion paper are measurements of those artifacts as they exist and are unaffected. The bug was fixed (identical init across ranks, explicit gradient all-reduce, checkpoint-time replica pinning) before every run in this paper; the frozen-kernel result reproduces identically under the corrected harness, and the accidental small-batch lineage becomes a free extra scale point in §3.

3 Position-kernel collapse

At every data scale — 9.5k items (1/8 recipe, 188 steps), 19.6k items (the 1/40 batch-4 lineage, 1,500 steps), and 76.0k items (full recipe, true batch 32, 1,500 steps) — the trained compactor's block-1 cross-attention rows are exact delta functions: mean row entropy at the numerical floor (1.16×10^{-10} nats, identical to five significant digits across init, all intermediate checkpoints, and all instances), and the argmax kernel — a uniform stride lattice, round($k\,(S-1)/1023$); stride ≈8 at the 8k training shape, stride ≈3 at the 4k deployment shape — **bitwise identical to initialization at every logged checkpoint of every run** (kept-set Jaccard 1.000 on every layer, between every pair of checkpoints and instances). Meanwhile the loss falls normally ($0.58 \to 0.36$ over 1,500 full-recipe steps) and the compactor's *behavior* improves monotonically: certified silent needle loss drops from 0.271 (1/40 scale) to 0.188 (full recipe) at matched budget, with the later instance's failure set nested inside the earlier's (94.1%). Every improvement is value-side learning over a frozen routing skeleton — the compactor learns what to write into fixed slots, never where to look. The loss curve is not merely uninformative about this; it is actively misleading, which is the operational point of §9.

Aggregation-axis certification is likewise scale-invariant under the frozen kernel: teacher KL 0.263 at 1/40 scale and 0.2566 at full recipe against eviction's 0.1218 (all n=12 documents) — an 8× increase in traced training data moved the certified aggregation number by 2%. Re-measured at n=60 with paired CIs (§6) the 1/40 and full-recipe rows are 0.2529 and 0.2820 against eviction's 0.0842: with 8× the traced data the certified aggregation number does not improve at all, strengthening the conclusion. What a frozen kernel can improve (value-side needle smuggling) improved; what it cannot (aggregation over surviving content) did not.

Two published-recipe implications follow. First, evaluation suites that aggregate over tasks would not distinguish this subsampler from a synthesizer; the two-axis protocol did (the subsampler's profile — aggregation-poor, needle-smuggling — is characteristic). Second, the recipe's own ablations imply full-scale published checkpoints route content; we could not reproduce escape by scale alone, which localizes the difference to configuration, not data (§5 scope note).

4 Mechanism: logit-scale-induced gradient saturation

One accumulated forward/backward at initialization over a fixed 128-item probe batch, gradient L_2 norms summed over layers by parameter group (Appendix A carries the full group×layer×scale table):

group	canonical ×256	$\times\sqrt{d_\ell} = 16$	ratio
cross-1 W_Q (+anchor)	1.1×10^{-11}	5.5×10^{-2}	5.0×10^{9}
cross-1 W_K (+anchor)	1.6×10^{-10}	5.4×10^{-1}	3.4×10^{9}
latent banks	6.1×10^{-2}	1.2×10^{-1}	1.9
cross-1 W_V	21.6	76.2	3.5
cross-1 W_O	21.6	76.2	3.5
block-2 cross (all)	21.6	76.2	3.5
self-attention (all)	17.3	61.1	3.5
output heads (W_{key}, W_{val})	21.6	76.2	3.5

Under the canonical scale the routing *projections* are numerically dead — eleven to twelve orders of magnitude below the value pathway, exactly along the routing/value cut. (The latent

banks show a small nonzero gradient at canonical scale because they also feed the residual stream and output heads; the clean routing-only weights are W_Q and W_K.) Swapping only the logit scale to $\sqrt{d_\ell}$, weights untouched, wakes precisely the dead groups by 10^9–10^{10} while value-path norms move a generic $\sim$3.5$\times$. The identity initialization produces near-one-hot attention rows; multiplied by 256, the softmax is saturated and passes $\sim$zero gradient to its logits; everything upstream of the attention weights starves, everything downstream trains.

Two controls sharpen the claim. Gaussian noise (σ=0.02) on the position anchors at canonical scale (arm E3) leaves the kernel frozen in its (perturbed) initial state — entropy pinned at the floor, kept-set Jaccard 1.000 across training, final loss indistinguishable from the canonical baseline (0.362 vs 0.365): degenerate tie-breaking is not the mechanism; saturation re-freezes whatever configuration it is handed. Content-seeded latent initialization at canonical scale (arm E4: per-head k-means centroids of corpus $[K; V]$ vectors, content projections un-zeroed) starts off-floor by construction (entropy 0.54 nats) and stays flat through training, with only layer 0's kept set drifting (mean cross-layer Jaccard 0.99), while paying a large penalty for breaking the identity passthrough (initial loss 5.89, final 0.586, and 2.6$\times$ the baseline's silent-loss rate on the fixed subsample): content in the pathway cannot substitute for gradient through it.

5 The cure and its dynamics

At $\sqrt{d_\ell}$ (arm E1), routing learns. The softer scale de-saturates the softmax at initialization (entropy starts at 0.815 nats rather than the floor), and training then *raises* it — flat until $\sim$step 75, rising smoothly to 1.11 nats by step 188 at mid scale, with the layer-0 argmax set expanding from 1,024 to $\sim$6,000 distinct source positions. The kernel moves: kept-set Jaccard vs initialization falls to 0.18 (mean over layers) by step 188. The movement is structured, not noise: the eight KV-heads **despecialize** from the single shared lattice (mean pairwise cross-head Jaccard 1.000 $\rightarrow$ 0.199) while remaining mostly item-invariant (cross-item Jaccard 0.821) — head-specialized, weakly content-conditioned routing at mid scale. At full recipe item-adaptivity deepens monotonically for 3,000 steps (cross-item Jaccard 0.789 $\rightarrow$ 0.550) while heads first scatter (cross-head $J \rightarrow 0.13$ by step 250) and then partially re-organize ($\rightarrow$ 0.29); the final kernel touches essentially every source position across the probe batch (8,186/8,188).

Training dynamics show the honest trade: the soft scale leaks the identity passthrough, so E1 *starts* worse (initial loss 0.737 vs the canonical 0.576; the $\times$256 scale was protecting the passthrough) — then crosses over before step 60 (15-step smoothed) and finishes well ahead: 0.219 vs 0.365 at mid scale, stable across three seeds (0.214/0.219/0.225). On the fixed 1k counterfactual subsample the escaped compactor also *loses fewer facts* at the same step count (y^+ 0.239 vs 0.370): escape does not trade short-horizon quality for mobility. The frozen configuration is not a bad optimum being escaped; it is a gradient-dead region the canonical constant walls in. A learned per-layer temperature initialized soft (arm E2, clamped $[1, d_\ell]$) escapes independently (final loss 0.2104, kernel Jaccard 0.172) and can recover sharpness where useful; we ship the fixed $\sqrt{d_\ell}$ per our pre-registered selection rule (final-loss tie $\Rightarrow$ simplest lever). We note the published recipe may differ from our reproduction in unexplored details; our collapse claim is scoped to the canonical configuration as specified, and the diagnostics of §9 resolve the question for any implementation in minutes.

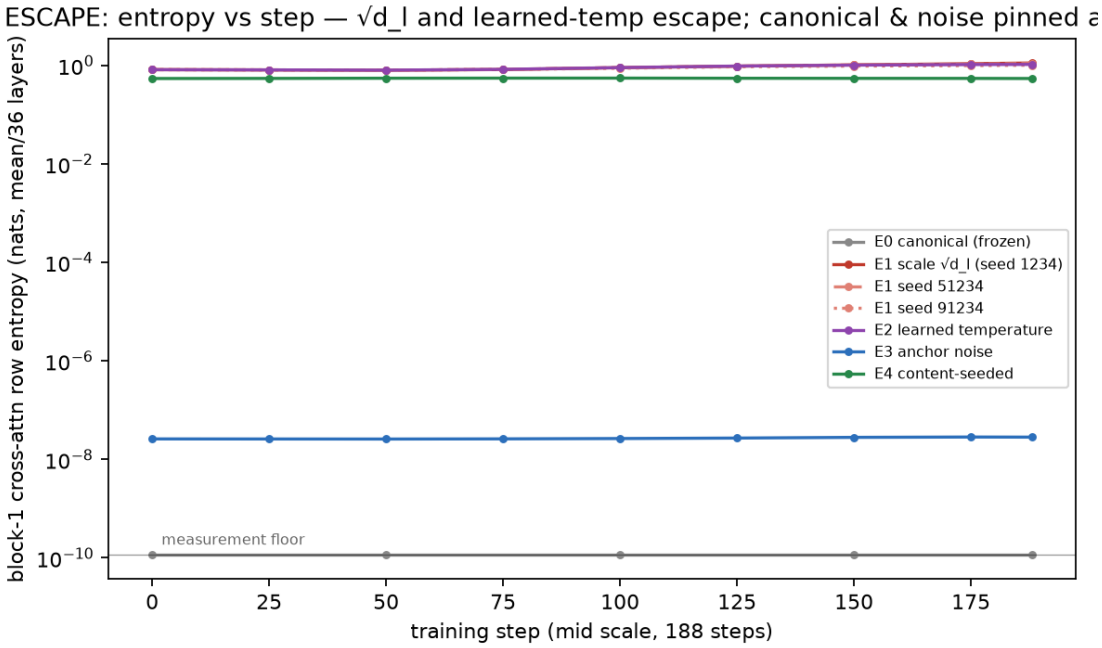


Figure 1: Entropy vs training step for every arm (log scale, mid-scale runs, 188 steps). Canonical (E0) and anchor-noise (E3) runs are pinned at the measurement floor; $\sqrt{d_\ell}$ (E1, three seeds) and learned temperature (E2) de-saturate at initialization and *rise* during training; content-seeded init (E4) is off-floor by construction but flat. Full-recipe trajectories in Figure 2.

6 Certification at full recipe

The historical n=60 table is invalidated: 52 evaluation documents and 207 probes appeared in full-recipe training. It therefore cannot establish either registered predicate, a frozen-versus-unfrozen ranking, or deployment eligibility. The replacement protocol reserves its 60 documents before manifest construction, binds holdout and training-manifest hashes into the checkpoint, records probe IDs, and fails on any overlap. Until that clean artifact exists, `max_certified_depth` = 0 and no compactor deployment is recommended.

7 Downstream: routing under a mobile kernel

Under the frozen lattice, the companion paper measured: write-time geometry loss-blind (needle-to-kept-distance AUC 0.509–0.517 at every scale and every intermediate checkpoint); a hidden-state probe ladder that reads behavioral divergence (0.854 subsampler / 0.802 frozen full recipe) while a context-only control sits at chance; and an expansion gate whose marginal silent-loss reduction declines monotonically as compressors improve (72% → 22.5% → 9.0% across eviction, 1/40, and frozen full recipe). At mid scale the escaped kernel did *not* flip the geometry result ($d_{\min}$ AUC ∼0.50 at every checkpoint, per-item kernels) — the strongest within-pilot test of the companion's internality claim.

The full-recipe re-interrogation sharpens all of it. **Geometry stays loss-blind even content-routed:** with per-item kept sets now rich (mean 2,779 positions vs the frozen 1,024

lattice), $d_{\min}$ AUC is 0.5003 $[0.495, 0.505]$ on the full 5,090-item protocol and ∼0.50 at all twelve checkpoints — the anticipated "geometry becomes legible when routing reads content" flip does not occur. Internality survives its strongest test: loss remains readable only from the computation, never from the write-time geometry, regardless of whether routing is positional or contentful. **The probe ladder recovers fully** on the unfrozen instance's labels (y^+ 0.487 → 0.325 → 0.173 across the three policies): stack 0.857 $[0.847, 0.865]$ vs the frozen dip 0.802 and the original 0.854; sink-heads 0.764; best read-out layer moves again (30 → 25 → 33) — probes are per-policy artifacts. The context-only null holds (0.507), and the two-hook sink-head monitor transfers zero-shot across the frozen↔unfrozen divide (0.759/0.727) — regime-general across compressor classes. **The gate's value reaches its floor:** refreshed on unfrozen labels (y^+ 0.083), the gate's optimal operating point is expand-nothing — marginal silent-loss reduction **0**%, completing the monotone trend 72% → 22.5% → 9.0% → 0%. Yet the computation-reading stack still routes where the gate cannot: it matches never-expand at 0.5% expansion and pushes below it (0.049 lost at 9.1% expansion). The refined ladder statement: as compressors improve, externally-visible routing signal is exhausted first; the computation-reading tier is the last to retain routing power, and at this generation it is the only one that does.

8 Fragility across the regime change

Within the frozen lineage, fragility nests: the full-recipe instance's failure set sits 94.1% inside the 1/40 instance's (κ 0.567–0.588 across scale, κ 0.745 across reseeds with nesting P=0.945). This work established that all frozen instances share one bitwise-identical kernel — so part of that stability could have been the shared lattice, a confound the companion paper's §7 could not have seen. The measurement across the frozen→unfrozen regime change resolves it in favor of the facts: per-instance loss rates fall generation over generation (0.499 → 0.418 → 0.321 → 0.172), cross-regime agreement drops (κ 0.28–0.42) — but **nesting largely survives**: 91% of the unfrozen instance's failure set lies inside the frozen 1/40 instance's, 78.5% inside the frozen full-recipe's. Much of the κ drop is marginal-driven (base rates 0.172 vs 0.499 cap κ even under perfect containment); genuinely novel losses are a minority (9–21.5%). The corrected claim — filed as an arXiv v2 note on the companion paper — is **narrower than we ourselves anticipated while drafting**: the shared frozen kernel accounts for the elevation from ∼0.8 to 0.94 nesting, not for the nesting phenomenon itself, which survives a kernel-class change largely intact. Fragility is largely content-intrinsic, with a real but minority regime-relative component. The per-policy rules stand regardless: gate labels, calibration, and probes are compaction-policy-scoped artifacts (the best-layer drift 30 → 25 → 33 alone guarantees this); a policy change invalidates the decision layer's training data along with the caches.

9 Diagnostics (the transferable deliverable)

Two checks, minutes each, that catch the collapse at step zero: **(1) init gradient forensics** — one forward/backward, per-group gradient norms; routing groups orders of magnitude below value groups means saturation. **(2) entropy telemetry** — cross-attention row entropy and argmax-set identity logged during training (ours costs one extra forward per 50 steps); floor-pinned entropy with a static kernel means the artifact is a subsampler regardless of its loss curve. Both are now mandatory in our certification protocol, and we recommend them for any

Perceiver-style compactor [1, 3], any resampler with sharpened attention, and any architecture combining identity-style initialization with large logit scales.

10 Limitations

One base model (4B); one compactor architecture and recipe family; escape levers swept at mid scale (188 steps) with the selected lever alone run to full recipe; the full-recipe unfrozen run is single-seed (mid-scale escape is three-seed; no within-regime κ exists yet for the unfrozen class); certification uses extractive-MCQ probes with letter-match equivalence, single-turn contexts, depths ≤ 4; published-recipe deltas beyond logit scale unexplored — the collapse claim is scoped to the canonical configuration as we implement it, and the §9 diagnostics are the portable test; the entropy floor (1.16×10^{-10}) is a bf16/softmax measurement limit, and sub-argmax mass was not telemetered (the frozen keepcount and loss-blind geometry make this unlikely to matter); cost figures are marginal rented-compute only.

11 Conclusion

A compactor whose loss falls for fifteen hundred steps at every data scale never learned the one thing compaction is for — where to look — because a single configuration constant walled its routing pathway behind a saturated softmax from the first gradient step. The loss curve could not see it; a one-pass gradient check and an entropy log see it immediately. One constant later, routing learns, heads specialize, the kernel reads content, and the unfrozen compactor beats the frozen one on the objective that froze it. Its certification result is withdrawn because the evaluation set overlapped training; a clean holdout rerun is required before any distributional or task-axis conclusion. The downstream measurements reorganize on cue, but not as a legibility story: geometry stays unreadable, the text-shadow gate's value falls to zero, and the only signal that keeps routing is the computation itself. The generalizable lesson is the method: certify per axis, instrument the pathway you care about — not the loss — and treat every improvement in the compressor as a fresh question about what still fails, and whether you can still see it.

References

[1] Vivek Chari, Guanghui Qin, and Benjamin Van Durme. KV-Distill: Nearly Lossless Learnable Context Compression for LLMs. *arXiv preprint arXiv:2503.10337*, 2025.

[2] memsub project. Reading the computation: How compression regime determines whether lost memory is routable. *preprint (companion paper, v0.4)*, 2026.

[3] Charles O'Neill, Alex Sandomirsky, Harry Partridge, Mudith Jayasekara, and Max Kirkby. Still: Amortized KV Cache Compaction in a Single Forward Pass. *arXiv preprint arXiv:2606.07878*, 2026.

A Gradient-forensics tables

Full group×layer norms at both scales: `memsub-artifacts:/escape/g1g2-gradient-table.json`. Aggregates in §4; init losses 0.705 (canonical) / 0.862 ($\sqrt{d_\ell}$; the passthrough-leak cost).

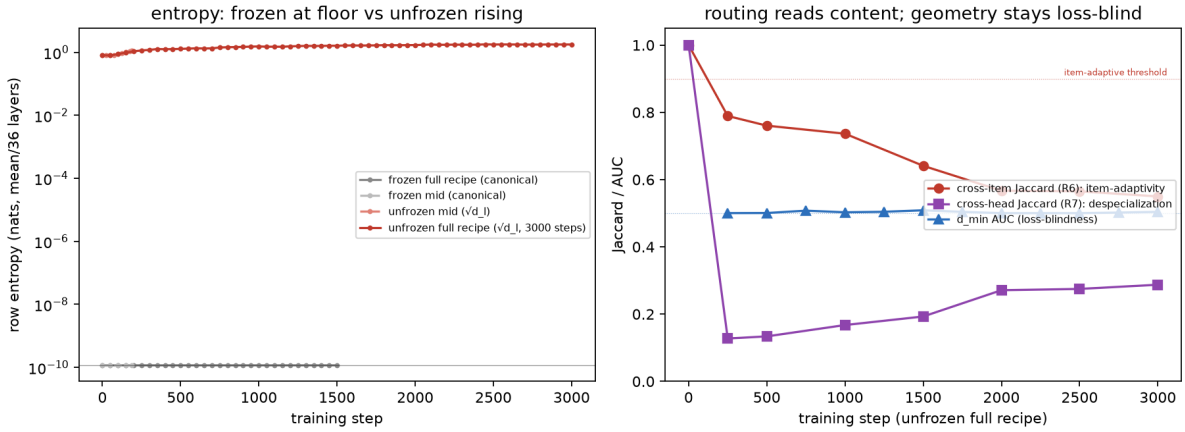


Figure 2: Full-recipe trajectories. Left: entropy — frozen runs pinned at the measurement floor; the unfrozen run rises to 1.78 nats and plateaus (the pre-registered extension to 3,000 steps fired at 1,500). Right: the kernel becomes item-adaptive (cross-item Jaccard $\rightarrow$ 0.550) and head-diverse while $d_{\min}$ stays at chance — routing reads content; geometry never becomes a loss signal.

B Trajectories

Entropy, kept-set Jaccard, cross-item (R6) and cross-head (R7) Jaccard, and per-checkpoint $d_{\min}$ AUC for all arms: `compactor-checkpoints/escape-*/telemetry/` and `fullrecipe-telemetry/i2-*`. Full-recipe trajectories: Figure 2 and `plans/260719-2328-e1full-unfrozen-full-recipe/`.

C Controls

E3 (anchor noise): floor-pinned, Jaccard 1.000, loss 0.362. E4 (content-seeded): entropy 0.539 flat, mean Jaccard 0.99 (layer-0 0.65), loss 5.89 $\rightarrow$ 0.586, subsample y^+ 0.614.

D Certification tables

Frozen lineage: `compactor-certification/summary-fullrecipe.json` (per-axis, noise floor 0.000, depth-4 collapse 15.8). Unfrozen: `compactor-certification/summary-e1full.json`.

E Downstream re-runs

Frozen-regime rows: arm table (`plans/260717-2319-.../arm-table.json`, `fullrecipe_*` rows), gate refresh (`gate-eval/gate-refresh-fullrecipe.json`). Unfrozen: `gate-eval/gate-refresh-e1ful` `probe/e1f-readside-*`, `e1full_*` arm-table rows, `q3-f1-verdict-e1full.json`, `q4-assembly-e1full.json`.

F Fragility matrices

Frozen-lineage: $\kappa(A{\leftrightarrow}B) = 0.745$ (P=0.945), $\kappa(A{\leftrightarrow}FR) = 0.567$, $\kappa(B{\leftrightarrow}FR) = 0.588$, nesting 94.1%. Cross-regime (E1F vs A / FR): κ 0.283/0.417, nesting 91.0%/78.5%, base rates

0.499/0.418/0.321/0.172 (`preprobe/e1-relabel-E1F.state.json`).

G Harness-provenance forensics

The M3.5 trainer invoked the compactor via `module.compact()`, bypassing the DDP wrapper; DDP arms its gradient hooks only in `forward()`, so gradients were never all-reduced (verified by a two-rank test: bypass ⇒ per-rank gradients differ; wrapper ⇒ identical). Shipped 1/40 instances are single-replica effective-batch-4 artifacts with per-rank initialization; rank 0's replica was saved. The corrected harness (identical init, explicit all-reduce, checkpoint-time replica broadcast) reproduces the frozen kernel bitwise at true batch 32 — the collapse is not a small-batch artifact.

H Reproducibility

Environment pinned (torch 2.7.1, transformers 4.55.2); training-mixture manifest sha256 `8e0bb280…`; policy versions embed weight hashes and certified depth. Marginal compute: collapse/full-recipe campaign ≈$440; escape pilot ≈$95; unfrozen full-recipe campaign ≈$390 (extension fired, as budgeted) — ≈$925 for the three campaigns of this paper. Cumulative project ≈$2,100.